Mathieu Padlewski

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Active metamaterials · Topological physics · Wave-based computing

Positions

ESPCI Paris, PSL University – Postdoc — Sep. 2026 – present
Institut Langevin
Project: *Time-Varying Media, Temporal Metamaterials and Wave Control*
Advisors: Prof. M. Fink, Prof. E. Fort

École Polytechnique Fédérale de Lausanne (EPFL) – Guest Scientist — Nov. 2025 – Aug. 2026
Laboratory of Wave Engineering (LWE)
Project: *Classical Emulation of Quantum Phenomena*

EPFL – Postdoc — Jul. 2025 – Oct. 2025
LWE (Collaboration with LG Electronics)
Project: *Active Noise Abatement of Electronic Appliances*
Advisors: Dr. S. Meyer, Dr. H. Lissek

EPFL – Research Intern — Mar. 2020 – Aug. 2021
Laboratory of Quantum Materials (QMAT)
Project: *Topological Semimetals under Torsional Strain*
Advisors: Dr. C. Putzke, Prof. P. Moll

Education

EPFL – PhD in Physics — Sep. 2021 – Jun. 2025
Laboratory of Wave Engineering (LWE)
Thesis: *Active Acoustic Metamaterials: A New Way to Understand Nonlinear and Topological Phenomena*
Advisors: Dr. H. Lissek, Prof. R. Fleury

EPFL – MSc in Physics — Sep. 2017 – Dec. 2019
Physics Laboratory of Complex Matter (LPMC)
Master's project: *Charge Density Waves and Transport Properties of 4Hb-TaS_2*
Advisors: Dr. E. Martino, Prof. L. Forró

University of Geneva – BSc in Physics — Sep. 2013 – Jun. 2017

Awards and Grants

Seal of Excellence – European Commission
Horizon Europe MSCA Postdoctoral Fellowship 2025, with CNRS
Project: *Wavebits: Analog Quantum-Inspired Computing*

Publications

Observation of amplitude-driven nonreciprocity for energy guiding — Mar. 2025
M. Padlewski, R. Fleury, H. Lissek, *Phys. Rev. B* **111**, 125156

Practical realization of chiral nonlinearity for strong topological protection — Jan. 2025
X. Guo, L. Jezequel, **M. Padlewski**, et al., *SciPost Phys.* **18**, 034

Active acoustic Su-Schrieffer-Heeger-like metamaterial — Jul. 2023
M. Padlewski, M. Volery, R. Fleury, et al., *Phys. Rev. Applied* **20**, 014022

Evidence of a coupled electron-phonon liquid in $NbGe_2$ Sep. 2021
H.-Y. Yang, X. Yao, V. Plisson, et al., *Nat. Commun.* **12**, 5292

Probing intraband excitations in $ZrTe_5$: A high-pressure infrared and transport study Mar. 2020
D. Santos-Cottin, **M. Padlewski**, E. Martino, et al., *Phys. Rev. B* **101**, 125205

Preprints

Classical analog emulation of quantum circuits via time-averaged dynamic states Jul. 2026
M. Valle, T. Tuuva, B. Apffel, H. Lissek, R. Fleury, **M. Padlewski**, arXiv:2601.02977

Hybrid broadband conduction and amplitude-driven topological confinement of sound via synthetic acoustic crystals Mar. 2026
M. Padlewski, X. Guo, M. Volery, R. Fleury, H. Lissek, arXiv:2408.16801

Talks and Conferences

IETR CNRS Institute, University of Rennes, France Jun. 2026
Invited talk

Wave Propagation and Control in Complex Media, Cargèse, France Apr. 2024
Selected conference talk

Forum Acusticum, Turin, Italy Sep. 2023
Conference talk

IRP Centre d'Acoustique Summer School, Sherbrooke, Canada Jun. 2023
Poster

Symposium on Acoustic Metamaterials, Nîmes, France Oct. 2022
Conference talk

Outreach

Listen to quantum atoms talk together thanks to acoustics
Dimensions – EPFL Magazine

Scientists use sound waves to unlock the secrets of quantum physics
The Brighter Side of News

Listen to quantum acoustic atoms – a new way to study the secrets of the universe
IFLScience – LabX Media Group

Des ondes sonores pour explorer la mécanique quantique
La Revue Polytechnique – Polymedia SA

The science is sound
Shaastra – IIT Madras

Peer Review

Physical Review Letters, Physical Review B, Communications Physics

Programming

Python, Qiskit, MATLAB, Simulink, LabVIEW, Mathematica, COMSOL, Inventor, EAGLE

Languages

French (native), English (native)

Activities

Music (composition, guitar); endurance sports; useless Arduino projects